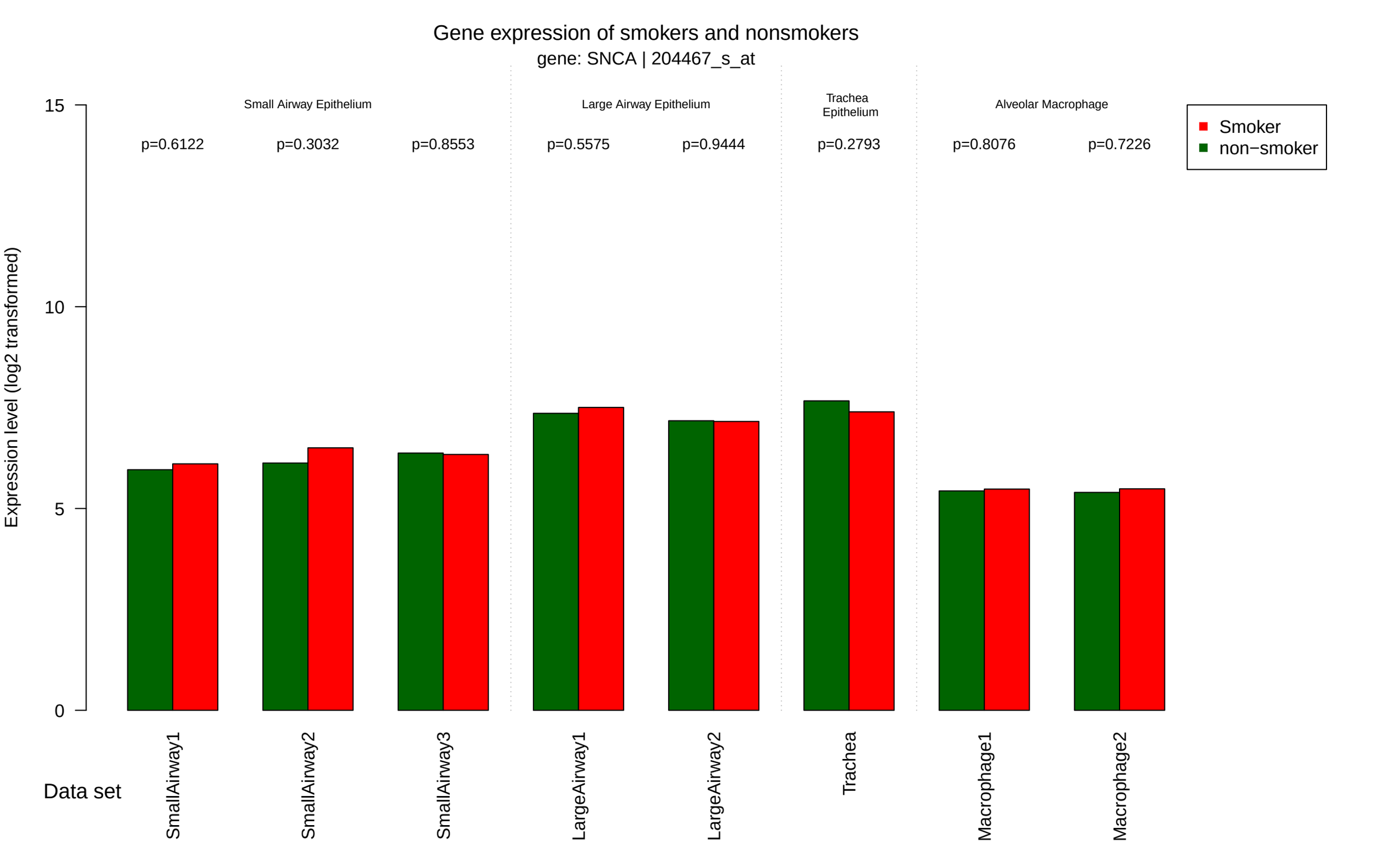
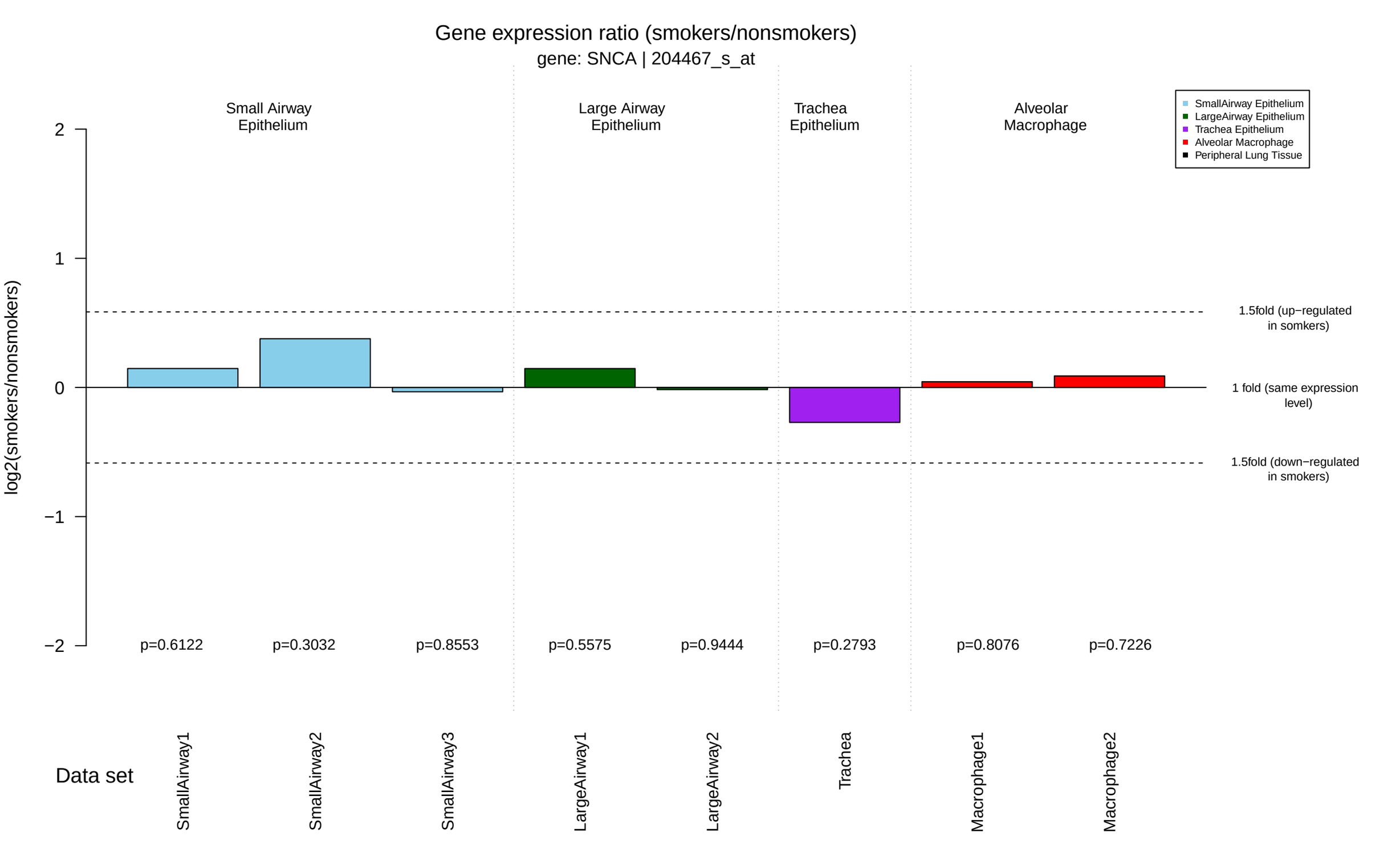
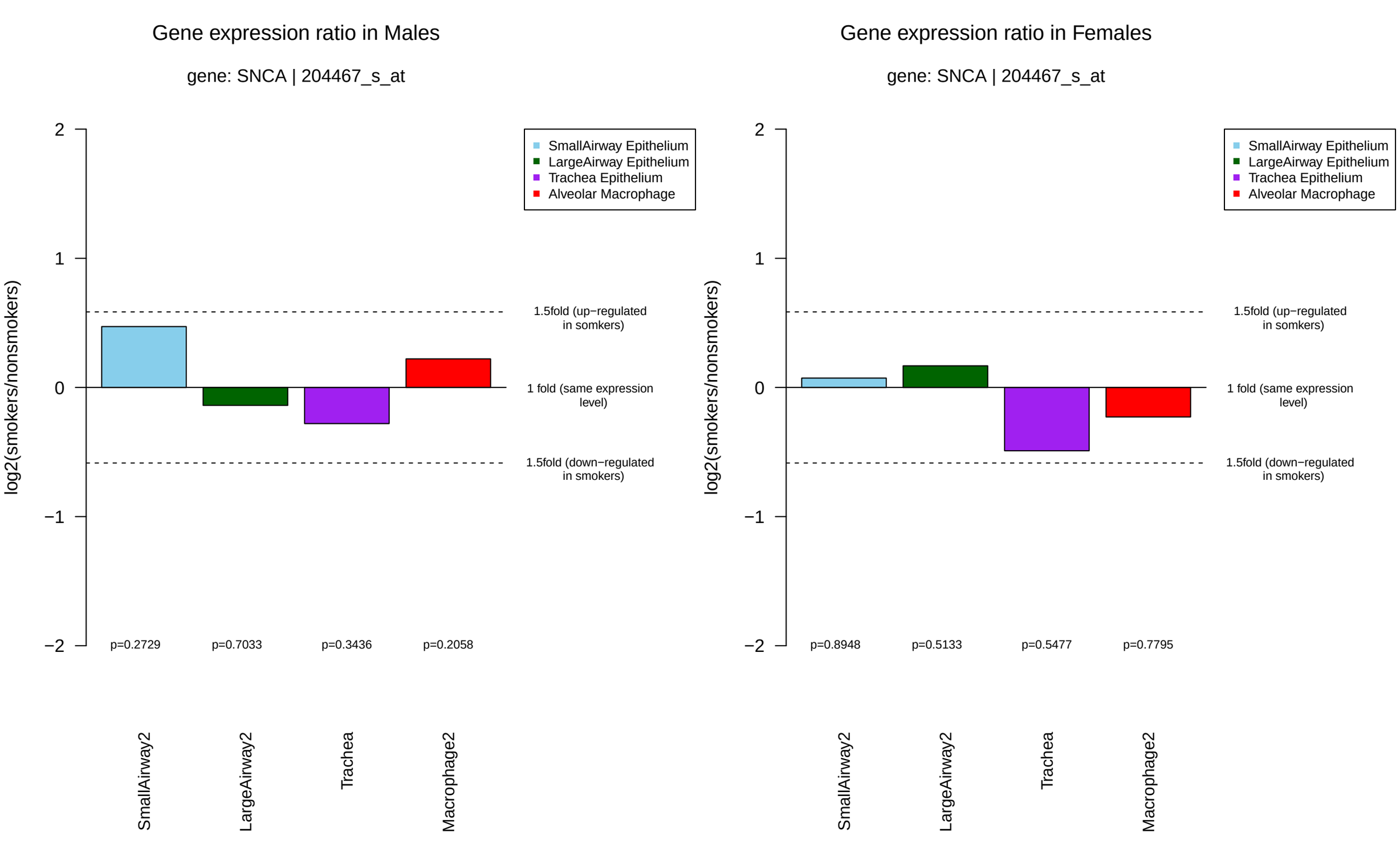
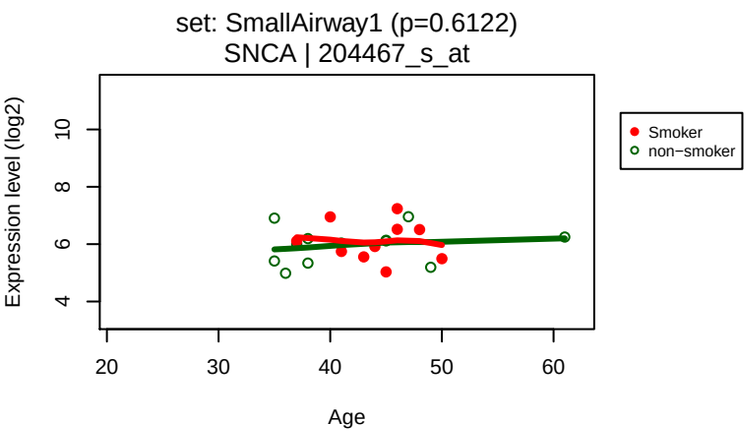
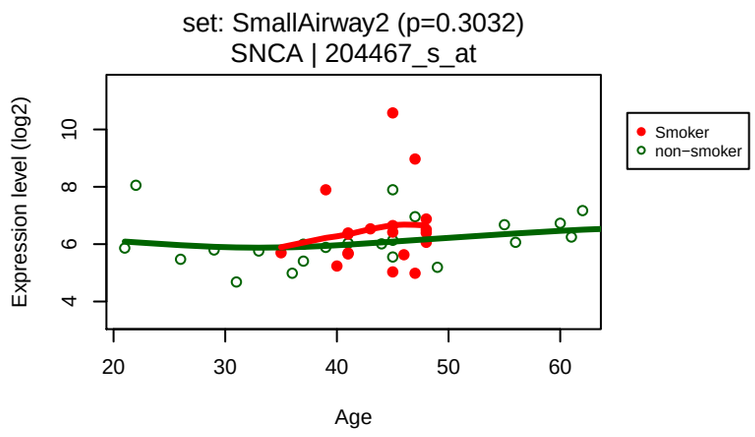
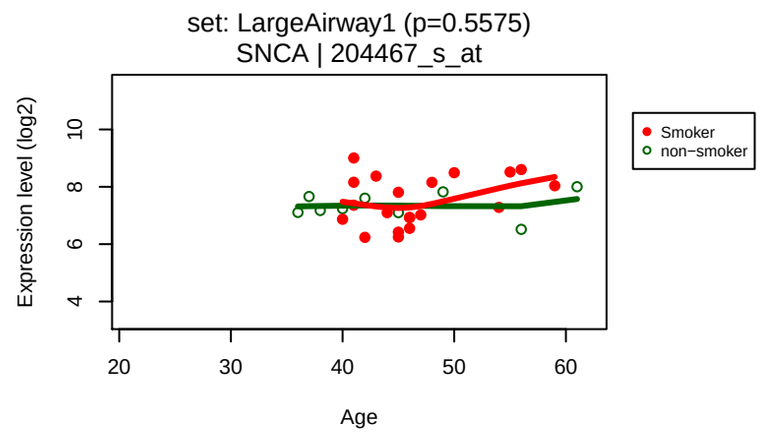

Gene expression of smokers and nonsmokers
gene: SNCA | 204467_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6122
p=0.3032
p=0.8553
p=0.5575
p=0.9444
p=0.2793
p=0.8076
p=0.7226
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

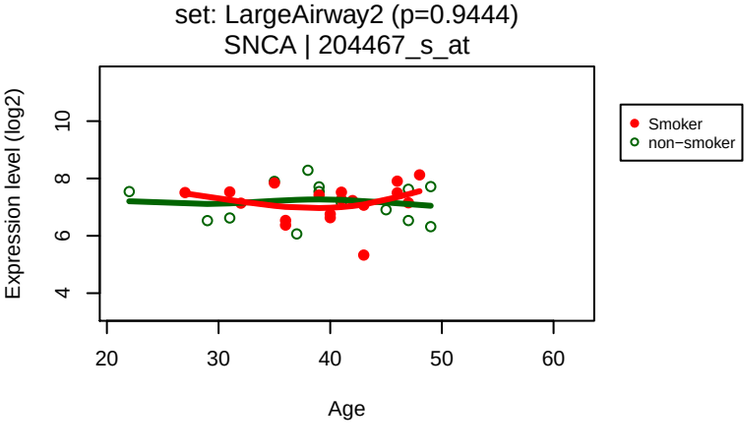

Gene expression ratio (smokers/nonsmokers)
gene: SNCA | 204467_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6122
p=0.3032
p=0.8553
p=0.5575
p=0.9444
p=0.2793
p=0.8076
p=0.7226
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

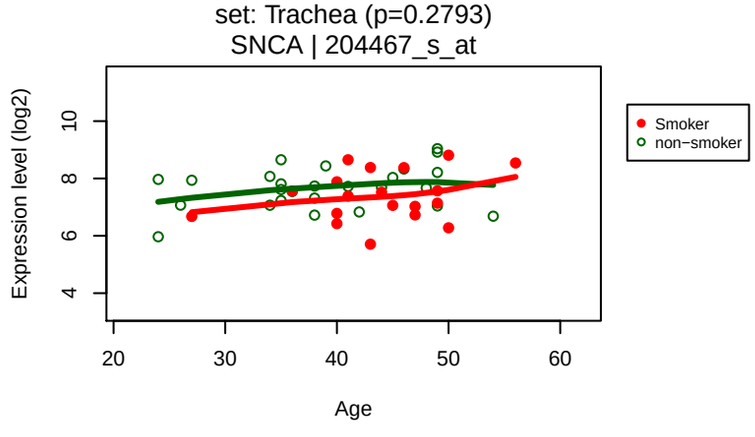

Gene expression ratio in Males
gene: SNCA | 204467_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2729
p=0.7033
p=0.3436
p=0.2058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SNCA | 204467_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8948
p=0.5133
p=0.5477
p=0.7795
SmallAirway2
LargeAirway2
Trachea
Macrophage2

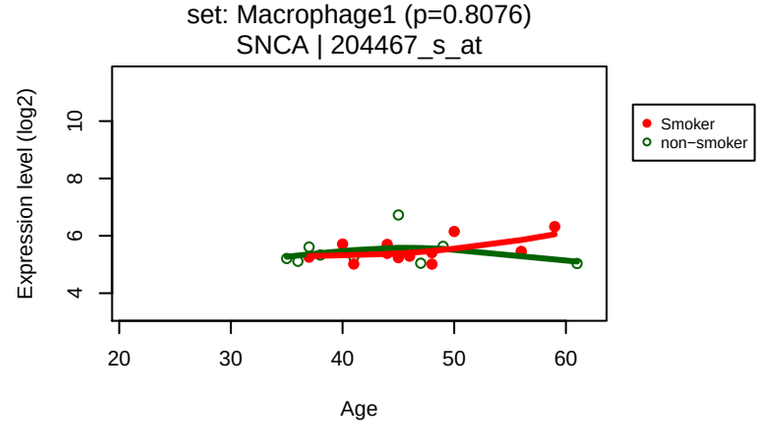

set: SmallAirway1 (p=0.6122)
SNCA | 204467_s_at
Expression level (log2)
Age
Smoker
non−smoker

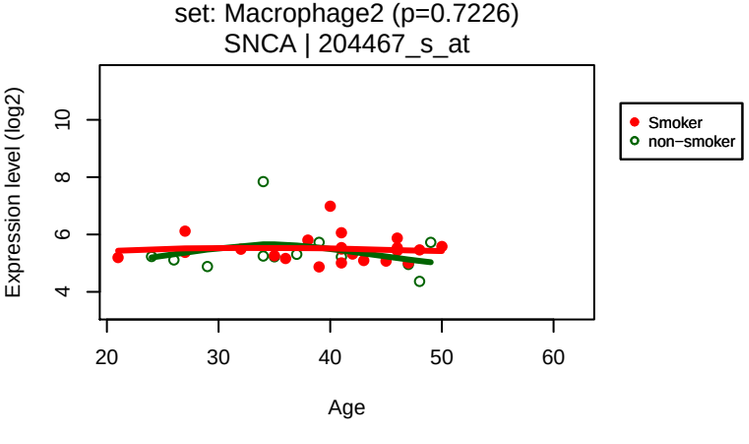

set: SmallAirway2 (p=0.3032)
SNCA | 204467_s_at
Expression level (log2)
Age
Smoker
non−smoker

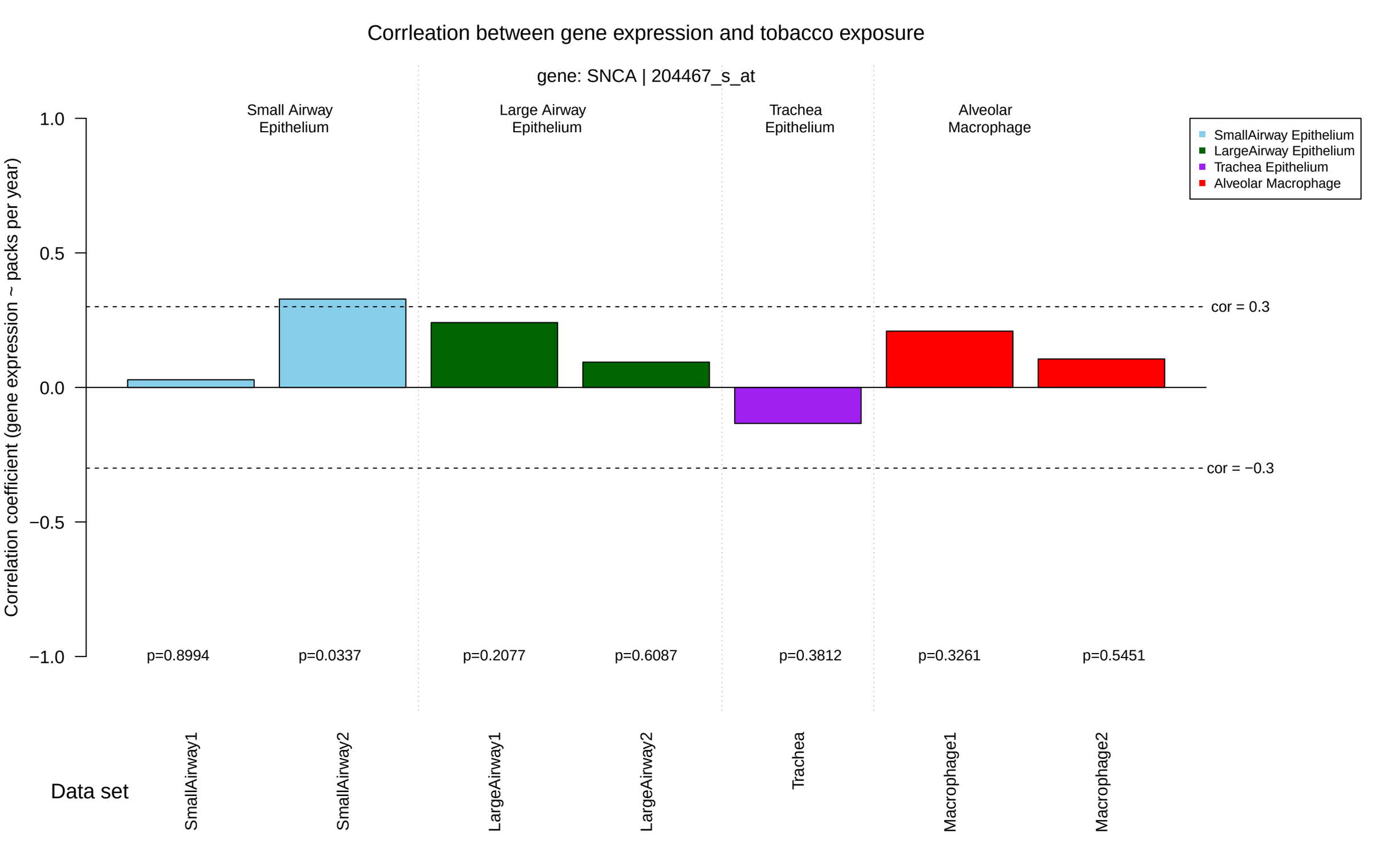

set: LargeAirway1 (p=0.5575)
SNCA | 204467_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9444)
SNCA | 204467_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2793)
SNCA | 204467_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8076)
SNCA | 204467_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7226)
SNCA | 204467_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SNCA | 204467_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8994
p=0.0337
p=0.2077
p=0.6087
p=0.3812
p=0.3261
p=0.5451
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2